Machine-Learning Model Implementation

1. Model Development

Following data preprocessing and feature selection, the final dataset contained **39,999 observations and 40 selected predictor variables**. The target variable was MENTHLTH, representing the number of mentally unhealthy days reported during the previous 30 days.

The dataset was randomly divided into **80% training data (31,999 observations)** and **20% testing data (8,000 observations)** using random_state = 42 for reproducibility.

Four conventional machine-learning models were evaluated:

- XGBoost
- Random Forest
- LightGBM
- Linear Regression

Tree-based models were trained using the original feature values, while predictors were standardized for Linear Regression.

2. Hyperparameter Tuning and Validation

Hyperparameters for XGBoost, Random Forest, and LightGBM were optimized using 5-fold cross-validation with Mean Absolute Error (MAE) as the selection criterion. The search grids are summarized below:

- XGBoost:

  n_estimators: [650, 700]

  max_depth: [2, 3]

  learning_rate: [0.02, 0.25]

  subsample: [0.7, 0.8]

- Random Forest:

  n_estimators: [700, 750]

  max_depth: [8, 10]

- LightGBM:

  n_estimators: [300, 500, 700],

  max_depth: [3, 5, 7, -1],

learning_rate: [0.02, 0.05, 0.1]

The best-performing hyperparameters for XGBoost were:

{'colsample_bytree': 0.7, 'learning_rate': 0.02, 'max_depth': 3, 'min_child_weight': 3, 'n_estimators': 700, 'reg_alpha': 0.1, 'reg_lambda': 0.1, 'subsample': 0.8}

All models were trained on the training set (31,999 rows) and evaluated on the hold-out test set (8,000 rows).

Model performance was assessed using:

- **MAE:** average absolute difference between observed and predicted mental-health days.
- **RMSE:** gives greater weight to larger prediction errors.
- **R^2:** proportion of variability in MENTHLTH explained by the model.

3. Model Performance

Model	CV MAE (mean ± std)	Test MAE	Test RMSE	Test R^2
XGBoost	4.100 ± 0.045	**4.142**	**6.789**	**0.342**
Random Forest	4.146 ± 0.052	4.183	6.859	0.328
LightGBM	4.151 ± 0.048	4.200	6.842	0.332
Linear Regression	4.318 ± 0.061	4.357	6.889	0.322

Among these models, XGBoost provided the best overall performance, with a test MAE of 4.14 days and an R^2 of 0.342. This indicates that the XGBoost predictions differed from the observed MENTHLTH values by approximately 4.14 days on average and explained about 34.2% of the variability in the target.

4. Summary

Overall, nonlinear models performed better than Linear Regression, suggesting that the relationships between the BRFSS predictors and mentally unhealthy days are not entirely linear. XGBoost was the strongest conventional machine-learning model.